

A note on variable capital utilization

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1 Model

Variable capital utilization is an important pattern of business cycles fluctuations as underlined by section 1

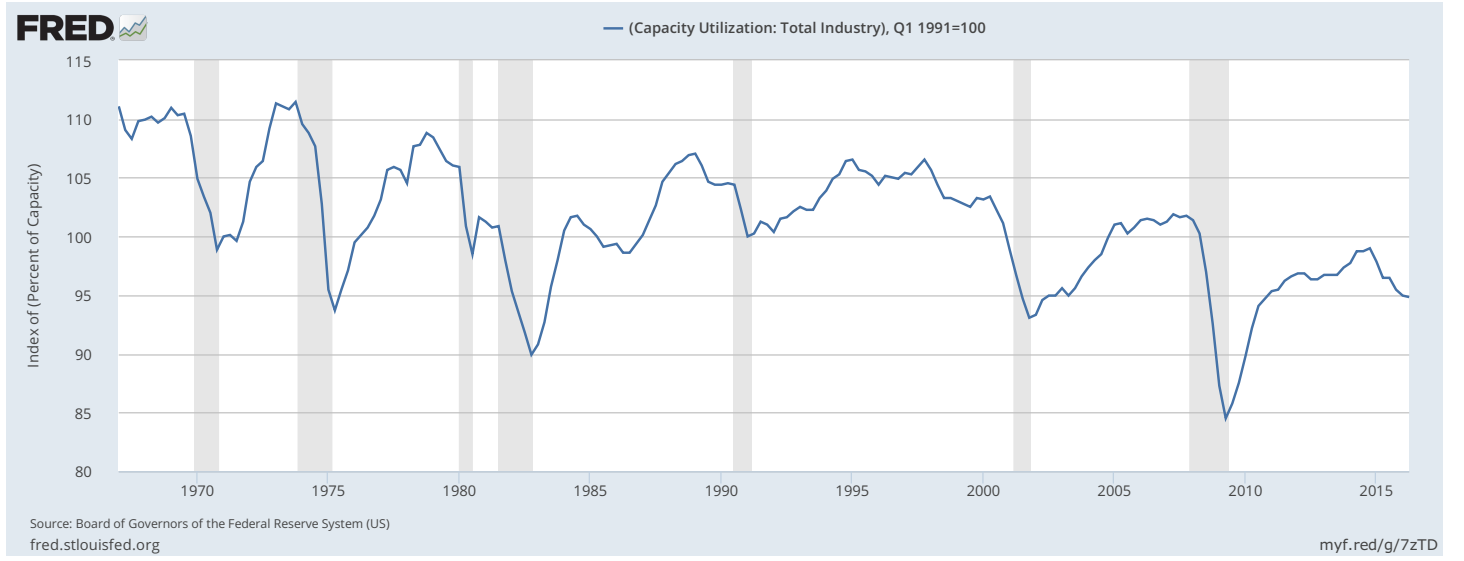


Figure 1: Capital utilization rate - US economy

Following Christiano et al. (2005), it is also possible to introduce variable capital utilisation in the production technology of firms:

$$Y_{it} = \varepsilon_t^A (K_{it}^u)^\alpha H_{it}^{1-\alpha}, \quad (1)$$

such that K_{it}^u denotes the capital utilised by the firms:

$$K_{it}^u = u_t K_{it-1}, \quad (2)$$

where u_t denotes the utilised level of capital with $\bar{u} = 1$, K_{it} is the amount of physical employed in the production process of goods. The utilisation rate may vary over the cycles: if $u_t > 1$, capital is over-utilised, which will force households to increase the capital supply, while if $u_t < 1$, capital is under-utilised.

Turning to households, they choose the optimal level of capital utilisation, u_{jt} . Over-utilisation of capital is costly for households in a proportion $\Phi(u_{jt})$:

$$\Phi(u_{jt}) = \bar{Z}(u_{jt} - 1) + \frac{1}{2} \frac{\psi}{1 - \psi} \bar{Z}(u_{jt} - 1)^2 \quad (3)$$

where $\bar{Z}$ is the steady state cost of capital services and ψ is the elasticity of capital utilisation. The budget constraint of households is now:

$$W_t H_{jt} + Z_t u_{jt} K_{jt-1} + R_{t-1} B_{jt-1} = C_{jt} + I_{jt} + \Phi(u_t) K_{t-1} + B_{jt} + T_{jt}, \quad (4)$$

Finally, the resource constraint is also affected by the utilisation cost paid in terms of goods:

$$Y_t = C_t + I_t + \bar{G} \varepsilon_t^G + \Phi(u_{jt}) K_{jt-1} \quad (5)$$

1.1 Households

Since Equation 4 is affected by capital utilization. We have to recalculate the first order conditions of the household optimization scheme. In addition, the capital utilization rate is optimally determined by households as they are the suppliers of physical capital to firms.

The Lagrangian problem is given by:

$$\begin{aligned}\mathcal{L}_t = & \mathbb{E}_t \sum_{\tau=0}^{\infty} \varepsilon_{t+\tau}^C \beta^\tau \left\{ \frac{C_{jt+\tau}^{1-\sigma_C}}{1-\sigma_C} - \chi \frac{H_{jt+\tau}^{1+\sigma_L}}{1+\sigma_L} \right. \\ & + \lambda_{t+\tau}^c [W_{t+\tau} H_{jt+\tau} + Z_{t+\tau} \textcolor{red}{u}_{jt+\tau} K_{jt-1+\tau} + R_{t-1+\tau} B_{jt-1+\tau} - C_{jt+\tau} - I_{jt+\tau} - B_{jt+\tau} - \textcolor{red}{\Phi}(u_{jt+\tau}) K_{jt-1+\tau} - T_{jt+\tau}] \\ & \left. + \lambda_{t+\tau}^k [\varepsilon_{t+\tau}^I I_{jt+\tau} + (1-\delta) K_{jt-1+\tau} - K_{jt+\tau}] \right\}\end{aligned}$$

If you're not comfortable with infinite sums, you can develop the sum for $\tau = 0$ and $\tau = 1$ (we don't mind for value of tau above 1):

$$\begin{aligned}\mathcal{L}_t = & \varepsilon_t^C \left(\frac{C_{jt}^{1-\sigma_C}}{1-\sigma_C} - \chi \frac{H_{jt}^{1+\sigma_L}}{1+\sigma_L} \right) \\ & + \varepsilon_t^C \lambda_t^c [W_t H_{jt} + Z_t u_{jt} K_{jt-1} + R_{t-1} B_{jt-1} - C_{jt} - I_{jt} - B_{jt} - \Phi(u_{jt}) K_{jt-1} - T_{jt}] \\ & + \varepsilon_t^C \lambda_t^k [\varepsilon_t^I I_{jt} + (1-\delta) K_{jt-1} - K_{jt}] \\ & + \varepsilon_{t+1}^C \beta \left(\frac{C_{jt+1}^{1-\sigma_C}}{1-\sigma_C} - \chi \frac{H_{jt+1}^{1+\sigma_L}}{1+\sigma_L} \right) \\ & + \varepsilon_{t+1}^C \lambda_{t+1}^c [W_{t+1} H_{jt+1} + Z_{t+1} u_{jt+1} K_{jt} + R_t B_{jt} - C_{jt+1} - I_{jt+1} - B_{jt+1} - \Phi(u_{jt+1}) K_{jt} - T_{jt+1}] \\ & + \varepsilon_{t+1}^C \lambda_{t+1}^k [\varepsilon_{t+1}^I I_{jt+1} + (1-\delta) K_{jt} - K_{jt+1}] \\ & + \varepsilon_{t+2}^C \beta^2 \left(\frac{C_{jt+2}^{1-\sigma_C}}{1-\sigma_C} - \chi \frac{H_{jt+2}^{1+\sigma_L}}{1+\sigma_L} \right) \\ & + \dots\end{aligned}$$

Only two first order conditions are different from what we had in the standard RBC model:

$$\begin{aligned}(\partial C_{jt}) : \lambda_t^c &= C_{jt}^{-\sigma_C} \\ (\partial H_{jt}) : \chi H_{jt}^{\sigma_L} &= \lambda_t^c W_t \\ (\partial B_{jt}) : \mathbb{E}_t \left\{ \frac{\varepsilon_t^C \lambda_t^c}{\varepsilon_{t+1}^C \lambda_{t+1}^c} \right\} &= \beta R_t \\ (\partial I_{jt}) : \lambda_t^c &= \varepsilon_t^I \lambda_t^k \\ (\partial K_{jt}) : \varepsilon_t^C \lambda_t^k &= \beta \mathbb{E}_t \left\{ \varepsilon_{t+1}^C \left[(1-\delta) \lambda_{t+1}^k + \lambda_{t+1}^c \textcolor{red}{u}_{jt+1} Z_{t+1} - \lambda_{t+1}^c \textcolor{red}{\Phi}(u_{jt+1}) \right] \right\} \\ (\partial u_{jt}) : \lambda_t^C \lambda_t^c [Z_t K_{jt-1} - \textcolor{red}{\Phi}'(u_{jt})] &= 0\end{aligned}$$

We can substitute the Lagrangian multipliers in (∂K_{jt}) by injecting (∂I_{jt}) and replace $\lambda_t^k = \lambda_t^c / \varepsilon_t^I$:

$$\begin{aligned}(\partial K_{jt}) : \varepsilon_t^C \lambda_t^k &= \beta \mathbb{E}_t \left\{ \varepsilon_{t+1}^C \left[(1-\delta) \lambda_{t+1}^k + \lambda_{t+1}^c \textcolor{red}{u}_{jt+1} Z_{t+1} - \lambda_{t+1}^c \textcolor{red}{\Phi}(u_{jt+1}) \right] \right\} \\ & : \varepsilon_t^C \frac{\lambda_t^c}{\varepsilon_t^I} = \beta \mathbb{E}_t \left\{ \varepsilon_{t+1}^C \left[(1-\delta) \frac{\lambda_{t+1}^c}{\varepsilon_{t+1}^I} + \lambda_{t+1}^c \textcolor{red}{u}_{jt+1} Z_{t+1} - \lambda_{t+1}^c \textcolor{red}{\Phi}(u_{jt+1}) \right] \right\} \\ & : \varepsilon_t^C \frac{\lambda_t^c}{\varepsilon_t^I} = \beta \mathbb{E}_t \left\{ \varepsilon_{t+1}^C \lambda_{t+1}^c \left[\frac{(1-\delta)}{\varepsilon_{t+1}^I} + \textcolor{red}{u}_{jt+1} Z_{t+1} - \textcolor{red}{\Phi}(u_{jt+1}) \right] \right\} \\ & : \frac{1}{\varepsilon_t^I} = \beta \mathbb{E}_t \left\{ \frac{\varepsilon_{t+1}^C}{\varepsilon_t^C} \frac{\lambda_{t+1}^c}{\lambda_t^c} \left[\frac{(1-\delta)}{\varepsilon_{t+1}^I} + \textcolor{red}{u}_{jt+1} Z_{t+1} - \textcolor{red}{\Phi}(u_{jt+1}) \right] \right\}\end{aligned}$$

In a second step, we use the first order condition (∂B_{jt}) to remove the Lagrangian multiplier on the budget constraint

λ_t^c :

$$\begin{aligned}
(\partial K_{jt}) : \frac{1}{\varepsilon_t^I} &= \beta \mathbb{E}_t \left\{ \frac{\varepsilon_{t+1}^C}{\varepsilon_t^C} \frac{\lambda_{t+1}^c}{\lambda_t^c} \left[\frac{(1-\delta)}{\varepsilon_{t+1}^I} + \textcolor{red}{u}_{jt+1} Z_{t+1} - \Phi(\textcolor{red}{u}_{jt+1}) \right] \right\} \\
: \frac{1}{\varepsilon_t^I} &= \frac{1}{R_t} \mathbb{E}_t \left\{ \left[\frac{(1-\delta)}{\varepsilon_{t+1}^I} + \textcolor{red}{u}_{jt+1} Z_{t+1} - \Phi(\textcolor{red}{u}_{jt+1}) \right] \right\} \\
: \frac{R_t}{\varepsilon_t^I} &= \mathbb{E}_t \left\{ \left[\frac{(1-\delta)}{\varepsilon_{t+1}^I} + \textcolor{red}{u}_{jt+1} Z_{t+1} - \Phi(\textcolor{red}{u}_{jt+1}) \right] \right\}
\end{aligned}$$

Regarding the first order condition on capital utilization, we have:

$$\begin{aligned}
(\partial u_{jt}) : \varepsilon_t^C \lambda_t^c [Z_t K_{jt-1} - \Phi'(u_{jt}) K_{jt-1}] &= 0 \\
(\partial u_{jt}) : Z_t - \Phi'(u_{jt}) &= 0 \\
(\partial u_{jt}) : Z_t &= \Phi'(u_{jt}) \\
(\partial u_{jt}) : Z_t &= \bar{Z} + \frac{\psi}{1-\psi} \bar{Z} (u_{jt} - 1)
\end{aligned}$$

as $\Phi'(u_{jt}) = \bar{Z} + \frac{\psi}{1-\psi} \bar{Z} (u_{jt} - 1)$.

1.2 Firms

The modifications are minor here to introduce variable capital utilization. We have to substitute K_{it-1} by K_{it}^u and create a new variable determining K_{it}^u :

$$K_{it}^u = u_t K_{it-1}$$

Equations where K_{it-1} was substituted by K_{it}^u are given by:

$$Y_{it} = \varepsilon_t^A (\textcolor{red}{K}_{it}^u)^\alpha H_{it}^{1-\alpha}, \quad (6)$$

$$Z_t = \alpha \frac{Y_{it}}{\textcolor{red}{K}_{it}^u} \quad (7)$$

1.3 Aggregation

The resource constraint is also affected:

$$Y_t = C_t + I_t + \bar{G} \varepsilon_t^G + \Phi(\textcolor{red}{u}_{jt}) \textcolor{red}{K}_{t-1}$$

2 Model summary

We have to include three new variables $\{\textcolor{red}{u}_t, \Phi(\textcolor{red}{u}_t), \textcolor{red}{K}_t^u\}$, and three corresponding equations to obtain a determinate system of equations:

$$\begin{aligned}
&: \beta R_t = \mathbb{E}_t \left\{ \frac{\varepsilon_t^C}{\varepsilon_{t+1}^C} \left(\frac{C_{jt+1}}{C_{jt}} \right)^{\sigma_C} \right\} \\
&: W_t = \chi H_t^{\sigma_L} C_t^{\sigma_C} \\
&: \frac{R_t}{\varepsilon_t^I} = \mathbb{E}_t \left\{ \frac{(1-\delta)}{\varepsilon_{t+1}^I} + Z_{t+1} \right\} \\
&: K_t = \varepsilon_t^I I_t + (1-\delta) K_{t-1} \\
&: \textcolor{red}{K}_t^u = u_t K_{t-1} \\
&: Z_t = \bar{Z} + (\psi/(1-\psi)) \bar{Z} (u_t - 1) \\
&: Y_t = \varepsilon_t^A (\textcolor{red}{K}_t^u)^\alpha H_t^{1-\alpha}, \\
&: Z_t = \alpha \frac{Y_t}{\textcolor{red}{K}_t^u} \\
&: W_t = (1-\alpha) \frac{Y_t}{H_t} \\
&: Y_t = C_t + I_t + g^y \bar{Y} \varepsilon_t^G + \Phi(u_t) K_{t-1} \\
&: \Phi(u_t) = \bar{Z} (u_t - 1) + (1/2) (\psi/(1-\psi)) \bar{Z} (u_t - 1)^2 \\
&: \log(\varepsilon_t^A) = \rho_A \log(\varepsilon_{t-1}^A) + \eta_t^A \\
&: \log(\varepsilon_t^G) = \rho_G \log(\varepsilon_{t-1}^G) + \eta_t^G \\
&: \log(\varepsilon_t^C) = \rho_C \log(\varepsilon_{t-1}^C) + \eta_t^C \\
&: \log(\varepsilon_t^I) = \rho_I \log(\varepsilon_{t-1}^I) + \eta_t^I
\end{aligned}$$

3 Steady state

The usual assumption regarding variable capital utilization is to normalize the steady state rate of utilization $\bar{u} = 1$, the steady state for $\{u_t, \Phi(u_t), \textcolor{red}{K}_t^u\}$ boils down to:

$$\begin{aligned}
\bar{u} &= 1 \\
\Phi(\bar{u}) &= 0 \\
\bar{K}^u &= \bar{K}
\end{aligned}$$

4 Dynare writing

1. Declare three new endogenous variables $\{u_t, \Phi(u_t), \textcolor{red}{K}_t^u\}$
2. Declare the elasticity of the utilization rate ψ as a parameter and calibrate it to $\psi=0.7$
3. Include the three dynamic equations defining $\{u_t, \Phi(u_t), \textcolor{red}{K}_t^u\}$ and modify equations affected by capital utilization.
4. Define the steady state of $\{u_t, \Phi(u_t), \textcolor{red}{K}_t^u\}$ in the init block.